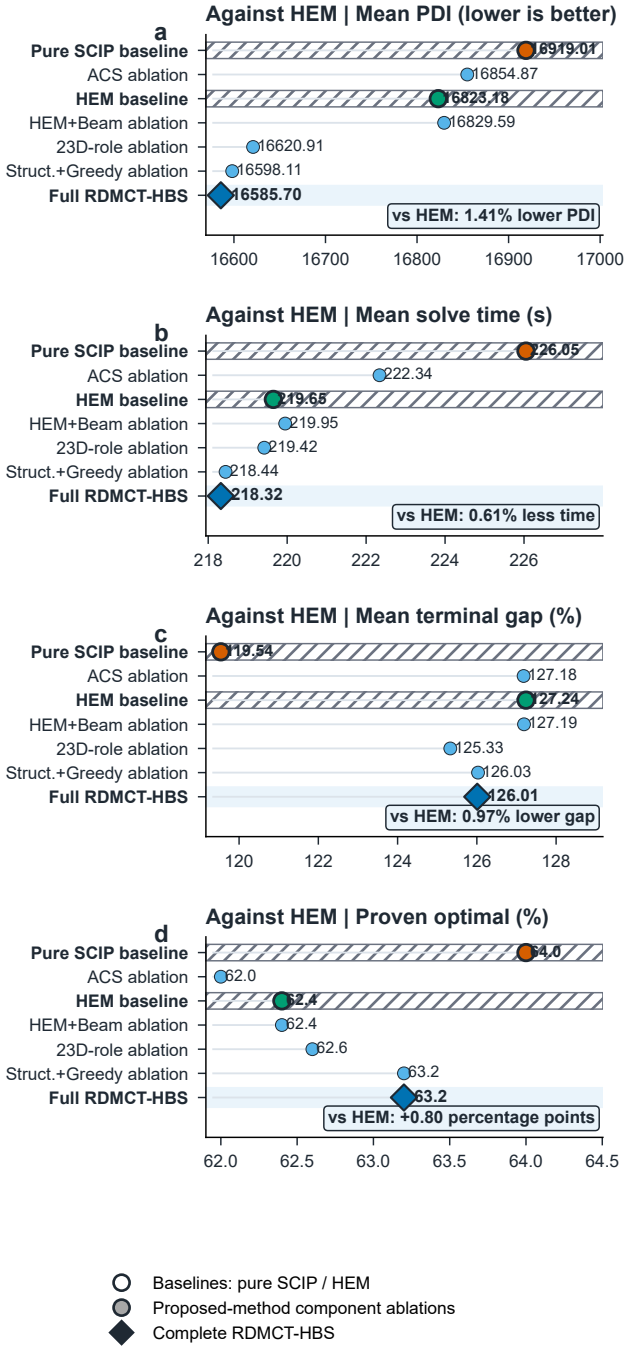

a
Against HEM | Mean PDI (lower is better)
Pure SCIP baseline
16919.01
ACS ablation
16854.87
HEM baseline
16823.18
HEM+Beam ablation
16829.59
23D-role ablation
16620.91
Struct.+Greedy ablation
16598.11
Full RDMCT-HBS
16585.70
vs HEM: 1.41% lower PDI
16600
16700
16800
16900
17000
b
Against HEM | Mean solve time (s)
Pure SCIP baseline
226.05
ACS ablation
222.34
HEM baseline
219.65
HEM+Beam ablation
219.95
23D-role ablation
219.42
Struct.+Greedy ablation
218.44
Full RDMCT-HBS
218.32
vs HEM: 0.61% less time
218
220
222
224
226
c
Against HEM | Mean terminal gap (%)
Pure SCIP baseline
ACS ablation
127.18
HEM baseline
127.24
HEM+Beam ablation
127.19
23D-role ablation
125.33
Struct.+Greedy ablation
126.03
Full RDMCT-HBS
126.01
vs HEM: 0.97% lower gap
120
122
124
126
128
d
Against HEM | Proven optimal (%)
Pure SCIP baseline
64.0
ACS ablation
62.0
HEM baseline
62.4
HEM+Beam ablation
62.4
23D-role ablation
62.6
Struct.+Greedy ablation
63.2
Full RDMCT-HBS
63.2
vs HEM: +0.80 percentage points
62.0
62.5
63.0
63.5
64.0
64.5
Baselines: pure SCIP / HEM
Proposed-method component ablations
Complete RDMCT-HBS